Aircraft Detection Report: WJA1884

SkySentinel Detection System
Report Generated: December 06, 2025
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Executive Summary

This report documents the detection and analysis of WJA1884, a Boeing 737 MAX 8 aircraft, on December 06, 2025 at 16:58:06 PST. The aircraft was detected at a critically low altitude of 340 feet while positioned 25.2 kilometers from the monitoring station. Frequency analysis revealed significant low-frequency acoustic energy, demonstrating the impact of aircraft noise on wildlife habitats.

Detection Event Details

Temporal Information

- **Detection Date:** December 06, 2025
- **Detection Time:** 16:58:06 PST
- **Trigger ID:** trigger20251206165806
- **Recording File:** aircraft20251206165806_88dB.wav

Aircraft Identification

- **Flight Number:** WJA1884
- **Aircraft Type:** Boeing 737 MAX 8
- **Route:** CYEG → PHOG

Acoustic Measurements

- **Audible Sound Level:** 88.2 dB (measured at monitoring location)
- **Peak Infrasound:** 162.5 dB at 6.9 Hz
- **Low Bass Peak:** 152.6 dB at 22.7 Hz
- **Infrasound vs Audible:** 74.3 dB difference

Aircraft Position Data

- **Altitude:** 340 feet AGL
- **Violation Classification: LOW ALTITUDE (distant)**
- **Distance from Monitor: 25.2 km (15.7 miles)**

- **Coordinates:** 47.8644°N, -122.7057°W

Frequency Analysis

Methodology

Audio recording analyzed using Fast Fourier Transform (FFT) to identify frequency components and their magnitudes across the audible and infrasound spectrum.

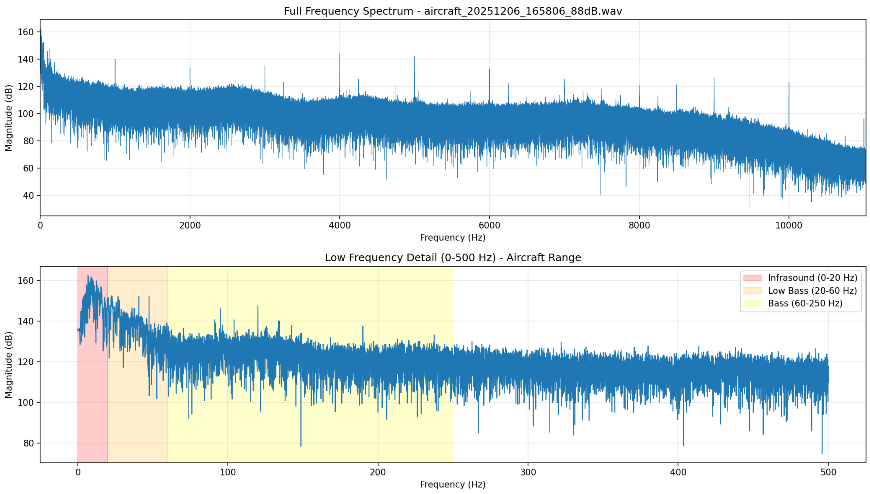

Figure 1: Frequency spectrum analysis showing aircraft acoustic signature

Frequency Band Results

Frequency Band	Peak dB	Peak Frequency	Average dB	Significance
Infrasound (0-20 Hz)	**162.5**	**6.9 Hz**	**147.9**	✈ STRONG
Low Bass (20-60 Hz)	**152.6**	**22.7 Hz**	**136.1**	✈ STRONG
Bass (60-250 Hz)	**147.5**	**120.0 Hz**	**122.4**	✈ STRONG
Low Mid (250-500 Hz)	**130.0**	**262.6 Hz**	**115.7**	
Mid (500-2000 Hz)	**140.3**	**1000.0 Hz**	**109.8**	
High Mid (2000-4000 Hz)	**144.5**	**4000.0 Hz**	**105.0**	
High (4000-8000 Hz)	**142.2**	**5000.0 Hz**	**98.3**	
Very High (8000+ Hz)	**126.3**	**8999.8 Hz**	**82.2**	

Regulatory Analysis

FAA Minimum Altitude Regulations

14 CFR § 91.119 - Minimum Safe Altitudes

The aircraft was operating at 340 feet AGL at a distance of 25.2 km from the monitoring location.

- **Measured Altitude:** 340 feet (below 500ft minimum)
- **Distance from Monitor:** 25.2 km
- **Assessment:** While the altitude is below FAA minimums, the aircraft was 25.2 km away and not directly over the monitoring location. This represents a **distant low-altitude operation** rather than a direct violation of the monitoring site's airspace.
- **Impact:** Despite distance, the aircraft produced significant infrasound detectable at the monitoring location, demonstrating far-reaching acoustic impact on wildlife habitat.

Wildlife Protection Considerations

Bald and Golden Eagle Protection Act (16 U.S.C. 668-668d)

The monitoring location is within documented bald eagle habitat. Low-altitude aircraft operations producing high-intensity infrasound may constitute disturbance to protected species.

Scientific Significance

This detection contributes to the growing body of evidence documenting:

- Aircraft infrasound propagation distances
- Wildlife exposure to low-frequency noise pollution
- Correlation between specific aircraft and acoustic signatures
- Need for expanded wildlife protection zones

Data Availability

Audio Recording: WJA188420251206audio.wav
Frequency Spectrum: WJA188420251206spectrum.png
System Logs: Available upon request

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System: SkySentinel v2.0 (Patent Pending)
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